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LECTURE 13

Mini-Projects

Twenty-three projects across the ten exercise sets of Part 2, in pairs: one forward on richer physics, one inverse from committed readings. Each takes its exercise further, and none has its answer written down in advance.

Deep Learning for Engineering · Aalborg University

One project per team of two or three · chosen after L12.2

What a Mini-Project Is, and Is Not

WHAT MAKES IT DIFFERENT FROM AN EXERCISE

- the answer is not known in advance, to you or to me
- you choose what to model and what to leave out, and defend the choice
- a negative result is a result, provided you can say why
- and the deliverable argues rather than reports

WHAT EVERY PROJECT NEEDS

a question	stated before you start
a baseline	classical, and taken seriously
a held-out test	chosen to be hard, not convenient
a limit	where your method stops working

THE BASELINE IS NOT OPTIONAL

Part 2 named cases where a classical method wins. Say whether this is one.

TEAMS OF TWO OR THREE

One project per team. Come and talk before you commit to one.

Heat Transfer

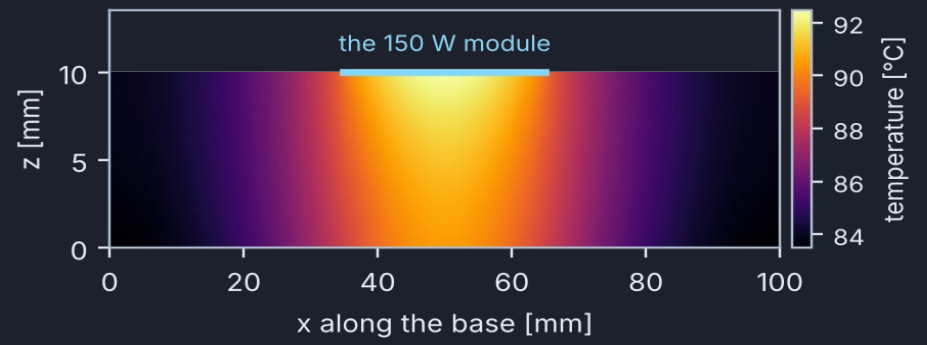
WHERE THE NETWORK BEATS THE CLASSICAL METHOD

identification	h_{eff} from one thermocouple, 5 % low
late transient	within 0.06 K at 150 s
built-in physics	hand calculation and balance in the loss
reuse	trained once, each answer under a ms

WHERE THE CLASSICAL METHOD STAYS AHEAD

One forward solve: finite differences meet 0.01 K in 4 ms steady, 0.05 K in 3 s for the pulse. The network's case is the inverse problem and the many-question study, not one solve.

THE FIELD THE NETWORK LEARNS



the temperature of the heat-sink base under the 150 W module, hottest point 92.45 °C

1 thermocouple

beside the module identifies the film coefficient of the whole fin array, 5 % low (Ex08, part 2)

Fluid Flow

WHERE THE NETWORK BEATS THE CLASSICAL METHOD

accuracy	2 to 6·10^{-6} of the steam fraction
against any grid	25 to 120× more exact than 401 × 41
design by gradient	Newton finds 0.3174 m/s, exact
many dimensions	the grid pays ×4 per halving of h

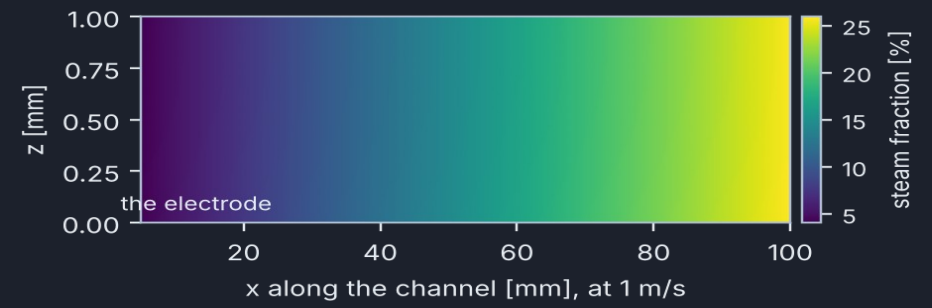
WHERE THE CLASSICAL METHOD STAYS AHEAD

One small solve: the pipe's grid meets 10^{-3} in 1 ms, and a stored field can be interpolated. The network pays with many queries, a speed as an input, and higher dimensions.

THE FIELDS THE NETWORKS LEARN



the velocity in the tube over the 4 s after the pump starts (Ex09.1)



the steam fraction along the fuel channel at 1 m/s; the network beats any grid tested (Ex09.2)

Electrochemistry

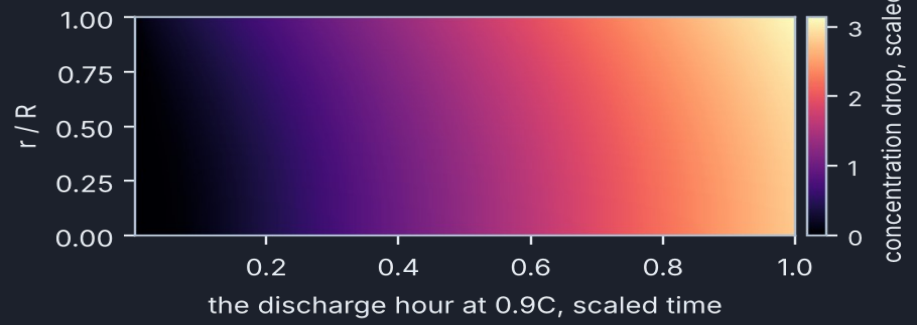
WHERE THE NETWORK BEATS THE CLASSICAL METHOD

one network	position, voltage, inlet temperature in
accuracy	0.19 K against the reference's 0.40 K
a thousand points	about 500× faster, same CPU
optimise through it	the cheapest hour by gradient
the particle	diffusivity back-solved, 0.30 % off

WHERE THE CLASSICAL METHOD STAYS AHEAD

For one operating point the coupled finite-volume solver is still the simple choice. The network's case is the whole operating window, carried by one training of 33 minutes.

THE FIELD THE NETWORK LEARNS



the drop of lithium concentration in the particle along the radius, over the discharge hour at 0.9C (Ex10.1)

500×

faster over a thousand operating points of the electrolysis cell, 0.19 K against the 50-cell reference's 0.40 K (Ex10.2)

Power Grid

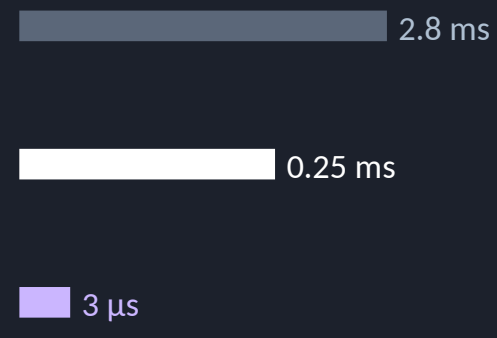
WHERE THE NETWORK BEATS THE CLASSICAL METHOD

state estimation	least-squares accuracy in 0.25 ms
inertia	3.89 s found, true 4.00, both starts
screening	a verdict in 3 µs a case
new operating points	within 0.6 ms of the simulation

WHERE THE CLASSICAL METHOD STAYS AHEAD

An unseen topology was 57 ms off, on the unsafe side: the graph network orders the queue of cases, and every unseen topology still goes to the swing simulation.

COMPUTATION TIME, ONE CASE



time for one answer, logarithmic scale - measured on a laptop CPU; the edge board is the mini projects' to measure

1000×

the screening gain: 3 µs a verdict against 2.8 ms a simulated case

Vision and Navigation

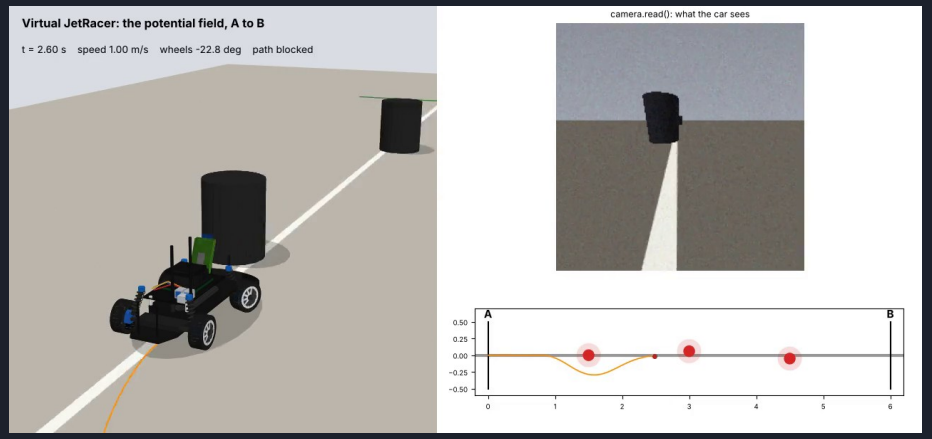
WHERE THE NETWORK BEATS THE CLASSICAL METHOD

the race	the rule crashes at 3.9 s; DL arrives
right frames	95.1 % against the dark rule's 85.1
false stops	none, against 13.5 % of free frames
the lap	within 6 cm up to 17 cm blind

WHERE THE CLASSICAL METHOD STAYS AHEAD

Nowhere on this task - but the grey box, as light as the floor, slips past rule and network alike. That is a limit of one camera, and the mini projects' second sensor answers it.

THE TWIN, DRIVING



the deep-learning JetRacer following the potential field around the obstacles, drawn by the exercise's own twin

8.49 s

A to B through the obstacles, five seeds alike - the conventional car crashes at 3.9 s (Ex11)

PINN Fundamentals

THE PROJECTS

MP7.1A	the slot with its insulation
MP7.1B	how old is the insulation?
MP7.2A	a die with a moving hot spot (optional set)
MP7.2B	where was the panel struck? (optional set)
MP7.2C	the eigenvalues of a real slot (optional set)

The pattern of every pair in Part 2: A forward on richer physics than the set, B inverse from committed readings.

THE BAR, PER PROJECT

MP7.1A	hot spot within 0.1 K, 7 points
MP7.1B	the age, from the readings
MP7.2x	the optional Helmholtz set's three

GROUND TRUTH

Built by tools/miniprojects/ex071_truth.py and ex072_truth.py, committed privately; an inverse project's answers stay back for grading.

Heat Transfer

THE PROJECTS

MP8.1A	the fins resolved
MP8.1B	two modules on one sink
MP8.2A	thermal cycling
MP8.2B	heat capacity and coefficient together

All four on the heat sink of Ex08: resolved into its fins, doubled, pulsed in time, and identified from readings.

THE BAR, PER PROJECT

MP8.1A	hottest 0.1 K, field 0.2 K, heat 1 %
MP8.1B	hottest 0.1 K per gap, optimum 2 mm
MP8.2A	peaks within 0.3 K, settling exact
MP8.2B	both constants within 5 %

GROUND TRUTH

Finite volumes on 0.25 mm cells (ex08_truth.py), committed privately; MP8.2B's two constants stay back for grading.

Fluid Flow

THE PROJECTS

MP9.1A	one bend of the serpentine, in 3-D
MP9.1B	the pump's speed reference
MP9.2A	the SOEC steam channel
MP9.2B	the air channel

9.1 is the liquid serpentine - the bend's solver is yours to find, and 9.1B's second case turns turbulent. 9.2 is the cell's gas channels.

THE BAR, PER PROJECT

MP9.1A	the solver proved 3 ways, then PINN
MP9.1B	module at most 80.2 °C, pumping 5 %
MP9.2A	speed 1 %, electrode 0.1 point
MP9.2B	electrode 0.1 point, Stefan answered

GROUND TRUTH

ex091_truth.py (with the Dean solver) and ex092_truth.py, committed privately; MP9.1A deliberately ships no 3-D reference.

Electrochemistry

THE PROJECTS

MP10.1A	the full model, where the SPMe stops
MP10.1B	how much has the cell aged?
MP10.2A	the channel across its height
MP10.2B	a day under an hourly price

10.1 is the battery particle: the full model where the single-particle model stops, and the cell's age from its charging history. 10.2 is the electrolyser: the electrode resolved, and a day of operation.

THE BAR, PER PROJECT

MP10.1A	voltage 20 mV at 1C, 3C; 50 at 5C
MP10.1B	both factors 5 %, capacity 1 %
MP10.2A	current 0.5 %, outlet mean 0.005
MP10.2B	the day's cost within 0.5 %

GROUND TRUTH

PyBaMM sets (Prada2013, Chen2020) and ex101_truth.py / ex102_truth.py, committed privately; the ageing factors stay back for grading.

Vision and Navigation

THE PROJECTS

MP11A	faster from A to B, on the JetRacer
MP11B	camera, LiDAR and ultrasonic in difficult conditions

The two hardware projects: both are developed on the virtual JetRacer first - pb.lidar, pb.sonar, pb.run_course - and graded on the real car, on the taped 6 m course with the lab's measured obstacle positions.

THE BAR, PER PROJECT

MP11A	no collision, at most 5.0 s (twin)
MP11B	5 % per kind, no collision (twin)
the car	five runs without contact, in the lab

GROUND TRUTH

ex11_truth.py: perfect perception's 8.76 s on 20 held-back courses, no frames saved; the RPLIDAR A1 and URM09 are ordered and in the lab.

Power Grid

THE PROJECTS

MP12.1A	the state of the 39-bus grid
MP12.1B	the inertia of every machine
MP12.2A	contingency screening on 39 buses
MP12.2B	the same screening with converters

The six-bus methods, unchanged, on the IEEE 39-bus grid - nothing harder. The solved optional sets, MP12.1A-39 to MP12.2B-39, are development areas beyond these.

THE BAR, PER PROJECT

MP12.1A	worst bus 0.01 p.u., beat WLS time
MP12.1B	inertias 5 %, dampings 20 %
MP12.2A	no miss, 5 % false, mean 5 ms
MP12.2B	no insecure case missed

GROUND TRUTH

ex121_truth.py and ex122_truth.py on grid39_classic.py: pandapower's case39 with the course's classical machine model, committed privately.

Choosing a Project

AN HONEST COMPARISON

fundamentals	the cleanest physics; the slot and the Helmholtz problems reward care, not compute
heat	easiest to start; the comb geometry and the double identification carry the interest
flow	most likely to produce a negative result honestly earned; 9.1A has no reference on purpose
electrochemistry	most data-driven; PyBaMM does the heavy lifting, so the question carries it
vision	most fun, most hardware risk; the twin first, then five runs on the real car
grid	most open-ended; the 39-bus grid is the one everyone publishes on

PICK BY THE QUESTION, NOT THE FIELD

The best projects come from wanting to know the answer. If none of these interests you, propose your own.

AND START WITH THE BASELINE

Every one of these has a classical method to beat, or to lose to honestly. Build that first.

ONE PROJECT PER TEAM OF TWO OR THREE

Chosen after L12.2; the chosen project's files come from the lecturer. Come and talk before you commit.